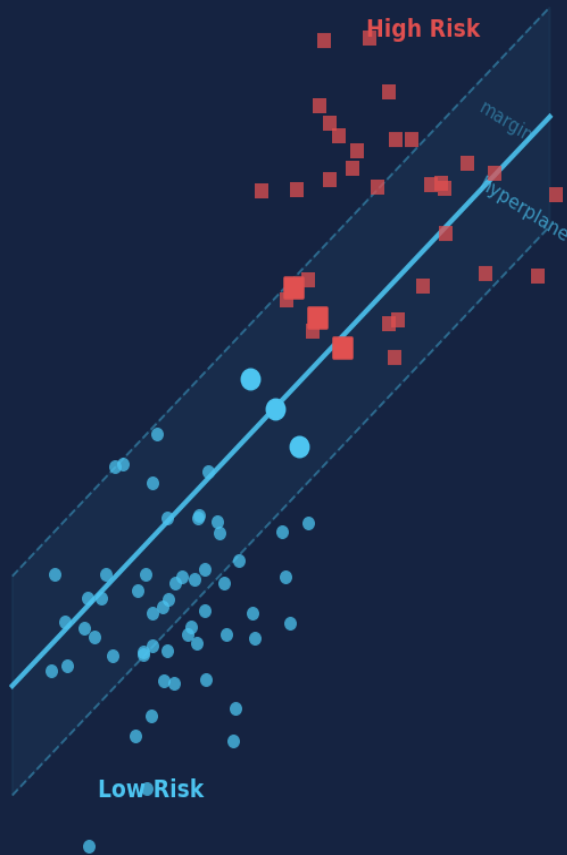


HR Employee Risk Analytics Are Not Random

What if disengagement could be detected before it becomes attrition?

Support Vector Machine · Risk Scoring · Early Warning



1,247

Employee Records

10

Features

68.9%

Accuracy

0.666

ROC-AUC

Disengagement, attrition, and performance decline rarely appear suddenly — they accumulate through measurable signals.



Attrition Cost

Replacing an employee costs
50–200% of their annual salary
in recruiting, onboarding, training



Late Detection

HR typically identifies risk
at resignation — after the signal
has been visible for months



Ops Impact

High-risk employees correlate
with scrap, quality failures
and team performance drag

30.7%

of employees show behavioral patterns consistent with high risk — detectable months before any formal event.

Old Thinking

Risk is reactive:

"The resignation happened."

Annual performance reviews.
Exit interviews after the fact.
No early signals. No prevention.
Just cost absorption.



New Thinking

Risk is probabilistic:

"How likely is this employee at risk?"

Score each employee quarterly.
Flag early signals before escalation.
Enable targeted HR conversations.
Turn behavioral data into decisions.

Three layers of employee data — behavioral, operational, and structural — each contributing a different dimension of risk:

BEHAVIORAL	Punctuality rate, Engagement score <i>Attendance patterns and self-reported engagement — the most direct disengagement signals</i>
OPERATIONAL	Productivity index, Scrap %, Training hours, Experience <i>Performance output and development investment — what the employee produces and receives</i>
STRUCTURAL	Department, Shift, Contract type, Area rotation rate <i>Organizational context — where the employee sits and under what conditions they operate</i>

Risk Rate by Structural Context

Contract	Temporary	40.6%	Permanent	25.7%	Outsourcing	34.2%
Department	Admin	41.6%	Production	37.4%	Quality	20.9%
Shift	Night	33.2%	Morning	32.3%	Afternoon	26.5%

Risk doesn't live in one variable — it emerges from specific combinations of behavior, performance, and context.

Employee risk doesn't follow a straight line. SVM finds the boundary that best separates risk patterns — even in non-linear feature space.

Non-linear Kernel (RBF)

Risk patterns combine multiple factors non-linearly. RBF kernel maps data into higher dimensions where a separating hyperplane exists — not possible with logistic regression.

Maximum Margin

SVM maximizes the margin between classes — it finds the boundary most likely to generalize. Especially valuable with limited labeled HR data.

Regularization via C

Parameter C balances fit vs. generalization. GridSearchCV found $C=50$, $\gamma='scale'$ — tuned on F1 to prioritize recall on the minority risk class.

Linear Companion

RBF SVM predicts. LinearSVC explains. Same data, same preprocessing — the linear coefficients give HR the 'why' behind each risk flag.

68.9%

Accuracy

0.666

ROC-AUC

49.4%

Precision (Risk)

45.8%

Recall (Risk)

LinearSVC Coefficients — Risk Drivers

shift_Night	+0.249
scrap_associated_pct	+0.170
engagement_score	-0.205
contract_Permanent	-0.210
experience_yrs	-0.130
training_hours_annual	-0.177

Confusion Matrix (312 test employees)

	Pred: Low Risk	Pred: High Risk
Actual: Low	171 (TN)	45 (FP)
Actual: High	52 (FN)	44 (TP)

52 high-risk employees missed — each one an undetected intervention opportunity.
Threshold tuning can improve recall.

Engagement -0.205

→ Strongest protective factor. Engagement survey scores below 2 trigger 1:1 HR conversation. This is the highest-leverage intervention available.

Permanent contract -0.210

→ Contract stability is protective — temporary and outsourced workers show 40.6% and 34.2% risk rates. Conversion to permanent is a structural lever.

Night shift $+0.249$

→ Night shift workers show 33.2% risk vs 26.5% afternoon. Consider rotation programs, shift incentives, and additional check-ins for night teams.

Scrap % $+0.170$

→ Scrap association signals operational stress or disengagement. Elevated scrap alongside low engagement is a compound flag — investigate both simultaneously.

Training hours -0.177

→ Investment in training is protective. Employees below 20 annual training hours are structurally underserved — prioritize them in development plans.

What-If Simulator — Employee Risk Scorecard

08

Same model. Three profiles. The third isolates the marginal effect of a targeted engagement intervention.

Employee A — Low Risk	Employee B — High Risk	Employee C — Intervention
Training50 hrs	Training5 hrs	Training5 hrs
Punctuality99%	Punctuality78%	Punctuality78%
Productivity110	Productivity80	Productivity80
Scrap2.0%	Scrap10.0%	Scrap10.0%
Engagement5/5	Engagement1/5	Engagement4/5 ↑
ContractPermanent	ContractOutsourcing	ContractOutsourcing
 28% risk probability	 100% risk probability	 99.6% risk probability

Scenario C insight: improving engagement alone (1→4) barely moves the needle from 100% to 99.6%. The risk pattern here is systemic — not a single-variable fix.

The score is a conversation starter — not a verdict. Context and human judgment determine the response.

This project is not about predicting who will leave.

It is about making disengagement visible while there's still time to act.

HR risk doesn't follow a linear rule. Risk emerges from combinations — and that's exactly why a kernel-based model captures it better.

Feel free to download the full project: dataset, notebook, and simulator.

Change the threshold. Adjust the kernel. Challenge what the model flags.

Because people development has no finish line.

